# Flatness of representation counts for prime sequences, I: the sub-peak spectrum and the second-order ripple

Kentaro Amauchi*

August 8, 2026

**Abstract**

Let $B$ be a finite set of odd primes and $r_B(m)$ the number of subsets of $B$ summing to $m$. The companion paper reduces the asymptotic behaviour of the subset-sum landscape of $B$ to a single quantity, the *flatness* $\varepsilon_d$, which measures the pointwise oscillation of $r_B$ on a bounded window. This paper develops the Fourier-side input needed to bound $\varepsilon_d$: we determine the complete spectrum of sub-peaks of the characteristic product $F_B$, we prove that modulus 6 is the unique maximiser of the per-element sub-peak height, and we show that for a truncated set of primes an entire family of sub-peaks vanishes *identically*. Concretely, the peak at modulus $q = 2m$ with $m$ odd is exactly 0 whenever $m \in B$; for $B$ a set of primes this kills every modulus $2p$ with $p \in B$, so that the only competitors of the modulus-6 peak are the modulus-4 floor $(1/\sqrt{2})^{|B|}$, which never vanishes, and a finite list depending on the truncation level. We also derive and verify numerically a second-order formula for the modulus-6 ripple, giving amplitude and phase to within 0.1% over $k = 18, \ldots, 40$. Key steps are formally verified in Lean 4.

## 1 Introduction

### 1.1 The quantity to be bounded

For a finite set $B$ of odd positive integers write $r_B(m) = \#\{S \subseteq B : \sum S = m\}$, $T = \sum B$, and

$$F_B(\theta) = \prod_{a \in B} (1 + e^{ia\theta}), \qquad r_B(m) = \frac{1}{2\pi} \int_0^{2\pi} F_B(\theta) e^{-im\theta}\, d\theta.$$

The companion paper shows that the ratio of the number of strict local minima to the number of ground states of the subset-sum landscape converges to an explicit arithmetic invariant provided the *flatness*

$$\varepsilon_d = \max_{m,m' \in \mathrm{Win}_d} \left| \frac{r_{B_d}(m)}{r_{B_d}(m')} - 1 \right|$$

tends to 0, where $B_d$ is the subsequence of elements exceeding $2d$ and $\mathrm{Win}_d$ is a bounded window about $T/2$. Bounding $\varepsilon_d$ is a circle-method problem: the main arc at $\theta \approx 0$ gives a smooth Gaussian term, and every deviation from smoothness is carried by the sub-peaks of $|F_B|$ at the rationals $\theta = 2\pi j/q$.

*Independent researcher. amkn.sub03@gmail.com, https://mathlab0911.github.io/

## 1.2 Results

Section 2 determines the sub-peak spectrum. Writing $G_B(\theta) = \prod_{a\in B}\cos(a\theta/2)$, so that $|F_B| = 2^b|G_B|$, where $b = |B|$, the natural per-element height at modulus $q$ is

$$M(q) = \Big(\prod_{\substack{1\le r\le q\\ \gcd(r,q)=1}} |\cos(\pi r/q)|\Big)^{1/\varphi(q)}.$$

Proposition 2.1 evaluates $M(q)$ in closed form, and Theorem 2.3 shows that $\sqrt{3}/2$ is its supremum, attained only at $q = 6$.

The novelty of Section 3 is that $M(q)$, being an upper bound valid for equidistributed sequences, is far from what a set of primes actually realises. Lemma 3.1 shows the peak at $q = 2m$ ($m$ odd) is *exactly zero* as soon as $m \in B$; Corollary 3.2 converts this into the surviving spectrum of a truncated prime set. The consequence for the circle method is that the minor arcs carry, up to a fixed finite list, only two live peaks.

Section 4 derives a second-order formula for the modulus-6 ripple, with amplitude correction $K_2$ and phase drift $\psi$; Section 5 treats the main arc and identifies the first correction to the central density, resolving an unexplained constant of the companion paper. Section 6 reports the numerical verification, Section 8 makes the finite list of remaining moduli effective, Section 9 treats the deep minor arcs, and Section 10 assembles the main theorem.

## 1.3 Relation to the literature

Three bodies of work are close enough to require comment.

*Oscillations in partition counts.* Murthy and Brack [MB26] give a semiclassical, periodic-orbit account of oscillations in partition counting functions; conceptually this is the same mechanism as ours, a few isolated saddle contributions producing a visible ripple. Their detailed oscillation analysis is for partitions into distinct *squares*, where the relevant orbits are indexed by Pythagorean triples, and their treatment of prime partitions (their §7) is asymptotic rather than oscillatory. The limit also differs: they let the integer being partitioned tend to infinity with the part set fixed, whereas we let the *number of parts* $k$ tend to infinity. The dictionary is one line: their periodic orbits are our sub-peaks at rational $\theta = 2\pi j/q$. We would add that the question they leave open — why regular oscillations survive only for certain special sequences — has, in the subset-sum setting, the exact answer given by Lemma 3.1 and Corollary 3.2: an oscillation at modulus $q$ survives precisely when $q$ avoids the cyclotomic vanishing $q = 2m$ with $m \in B$, and the surviving heights are the $M(q)$ of Proposition 2.1.

*Smoothness of partition functions.* Bateman and Erdős [BE56] prove $p_A(n-1)/p_A(n) \to 1$ for a *fixed infinite* set $A$ of parts with *repetitions allowed.* Our situation differs in both respects: our part set is a finite truncation that grows with $k$, and parts are used at most once, so the generating function is $\prod_{a\in B}(1 + x^a)$ rather than $\prod_{a\in A}(1 - x^a)^{-1}$. We do not see how to deduce the finite-truncation statement from theirs. The same remark applies to Roth and Szekeres [RS54] and to Vaughan [V08], both of which treat the fixed infinite set of all primes.

*Landscapes of the number partitioning problem.* Ferreira and Fontanari [FF98] compute the fraction of metastable states — that is, $\mathrm{lm}/2^k$ — analytically for uniform random weights; Alyahya and Rowe [AR14] give a formula in terms of the coefficient of variation; Stadler, Hordijk and Fontanari [SHF03] analyse barrier trees. All of these average over *random* weights, whereas the companion paper and this one treat a *specific* sequence and an order-sensitive invariant.

Finally, the anti-concentration literature (Littlewood–Offord and its inverse forms) bounds the *maximum* of $r_B$; what we need is a two-sided bound on the *ratio* at nearby points, which does not follow from a one-sided bound.

### 1.4 Formal verification

Statements marked `name` are machine-checked in Lean 4 against Mathlib; the repository builds with no `sorry` and no additional axioms. The canon currently consists of 97 theorems.

## 2 The sub-peak spectrum

The two statements of this section are classical: Proposition 2.1 is the resultant $\mathrm{Res}(\Phi_q, x+1)$ written out, and Lemma 2.2 is a standard exercise. We record both with proofs because we use the precise constants, and because the interpretation of $M(q)$ as the per-element height of a sub-peak of a subset-sum generating function seems not to have been made before.

**Proposition 2.1** (Cyclotomic evaluation; classical)**.** *For every integer $q \geq 3$,*

$$M(q) = \tfrac{1}{2}\,|\Phi_q(-1)|^{1/\varphi(q)},$$

*where $\Phi_q$ is the $q$-th cyclotomic polynomial.*

*Proof.* Put $\zeta = e^{2\pi i/q}$. For $1 \leq r \leq q-1$ we have $|1+\zeta^r| = 2|\cos(\pi r/q)|$. Since $\Phi_q(x) = \prod_{\gcd(r,q)=1}(x-\zeta^r)$, evaluating at $x=-1$ gives $\prod_{\gcd(r,q)=1}|1+\zeta^r| = |\Phi_q(-1)|$, hence $\prod_{\gcd(r,q)=1}|\cos(\pi r/q)| = 2^{-\varphi(q)}|\Phi_q(-1)|$. Take $\varphi(q)$-th roots. □

**Lemma 2.2** (Classical values)**.** *For $q \geq 3$ one has $\Phi_q(-1) = p$ if $q = 2p^j$ with $p$ an odd prime and $j \geq 1$; $\Phi_q(-1) = 2$ if $q = 2^j$ with $j \geq 2$; and $\Phi_q(-1) = 1$ otherwise.*

*Proof.* We use three standard identities: $\Phi_n(1) = p$ if $n = p^j$ is a prime power and $\Phi_n(1) = 1$ if $n$ has at least two distinct prime factors; for odd $n \geq 3$, $\Phi_{2n}(x) = \Phi_n(-x)$; and for even $m$, $\Phi_{2m}(x) = \Phi_m(x^2)$. If $q$ is odd then $\Phi_q(-1) = \Phi_{2q}(1) = 1$ because $2q$ has two distinct prime factors. If $q = 2m$ with $m$ odd then $\Phi_q(-1) = \Phi_m(1)$, which is $p$ when $m = p^j$ and 1 otherwise. If $4 \mid q$, write $q = 2m$ with $m$ even; then $\Phi_q(-1) = \Phi_m(1)$, which equals 2 exactly when $m$ is a power of 2, i.e. when $q = 2^j$, and equals 1 otherwise. □

**Theorem 2.3** (Extremality of modulus six)**.** $\sup_{q\geq 3} M(q) = \sqrt{3}/2$*, attained only at $q = 6$. The second largest value is $M(10) = 5^{1/4}/2 \approx 0.7477$, and $M(q) \leq 1/\sqrt{2}$ for every $q \notin \{6, 10\}$.*

*Proof.* By Proposition 2.1 and Lemma 2.2, $M(q) = 1/2$ unless $q = 2p^j$ or $q = 2^j$. For $q = 2^j$ ($j \geq 2$) we get $M = \frac{1}{2} \cdot 2^{1/2^{j-1}}$, which is decreasing in $j$ with maximum $1/\sqrt{2}$ at $q = 4$. For $q = 2p^j$ we get $M = \frac{1}{2}\,p^{1/(p^{j-1}(p-1))}$; the exponent decreases in $j$, so the maximum over the family occurs at $j = 1$, where $M = \frac{1}{2}p^{1/(p-1)}$. The map $p \mapsto (\log p)/(p-1)$ is strictly decreasing for $p \geq 3$, since its derivative has the sign of $(p-1)/p - \log p < 0$ (as $\log p \geq \log 3 > 1 > (p-1)/p$). Hence the values in this family are $\sqrt{3}/2 > 5^{1/4}/2 > 7^{1/6}/2 \approx 0.6916 > \cdots$, and for $j \geq 2$ the largest is $q = 18$ with $\frac{1}{2}3^{1/6} \approx 0.6005$. Combining the three families gives the statement. □

(Lean: `M12_six_maximal` and `M12_gap` verify the finite comparison at the twelfth power, where all the relevant values are rational.)

## 3 The vanishing family and the surviving spectrum

**Lemma 3.1** (Exact vanishing)**.** *Let $q = 2m$ with $m$ odd. If $m \in B$ then $F_B(e^{2\pi i/q}) = 0$; equivalently $G_B(2\pi/q) = 0$.*

*Proof.* The factor of $F_B$ indexed by $m$ is $1 + e^{i\pi} = 0$. ☐

(Lean: `F_eq_zero_of_mem` for the product form, `F_eq_zero_at_two_pi_div` for the statement at $\theta = 2\pi/q$, `F_nat_eq_zero_of_mem` for integer sequences, and `not_mem_of_F_ne_-zero` for the contrapositive. The special case $q = 6$, $m = 3$ is `normSq_at_pi` of the companion paper.)

The point of Lemma 3.1 is its converse for prime sequences: the residue class $a \equiv m \pmod{2m}$ consists of odd multiples of $m$, so if $B$ is a set of primes and $m > 1$ then $B$ meets that class if and only if $m$ itself lies in $B$. This gives a complete description.

**Corollary 3.2** (Surviving spectrum)**.** *Let $B_d$ be the set of primes in $(2d, x]$ and $b = |B_d|$. Then:*

1. $|G_{B_d}(2\pi/6)| = (\sqrt{3}/2)^b$ exactly*, provided $d \geq 2$ so that $3 \notin B_d$; no equidistribution is needed, because both classes $\pm 1 \pmod 6$ give $|\cos(\pi a/6)| = \sqrt{3}/2$.*

2. $|G_{B_d}(2\pi/4)| = (1/\sqrt{2})^b$ exactly*, for every odd sequence whatsoever, since $|\cos(\pi a/4)| = 1/\sqrt{2}$ for all odd $a$. This is the Fourier form of the modulus-4 obstruction of the companion paper.*

3. *$|G_{B_d}(2\pi/(2p))| = 0$ exactly, for every prime $p$ with $2d < p \leq x$. The vanishing is* pointwise *and must not be read as the absence of a peak: one factor is removed at one point, and the surrounding mass is still $(M(2p) + o(1))^b$. The two maxima flanking the resulting notch sit at height $(M(2p) + o(1))^b b^{-3/2}$, the notch having width $\asymp 1/p$ — see Remark 11.16, where this is measured and matched against Lemma 11.11.*

4. *For the finitely many primes $p \leq 2d$ the peak at $q = 2p$ survives, at height $(M(2p) + o(1))^b \leq (5^{1/4}/2 + o(1))^b$ by the prime number theorem in arithmetic progressions.*

5. *For $q = 2m$ with $m$ odd composite, and for $q = 2^j$ with $j \geq 3$, the height is $\leq (M(q) + o(1))^b \leq (\frac{1}{2} 3^{1/6} + o(1))^b$.*

6. *For $q$ odd the height is $\leq (\frac{1}{2} + o(1))^b$, and at $\theta = \pi$ it is $0$ by parity.*

*Remark* 3.3. Items (i)–(iii) and (vi) are exact identities; only (iv)–(v) use equidistribution, and only for a finite list of moduli, for which effective error terms in the prime number theorem in arithmetic progressions are available. The practical consequence is that the minor-arc analysis no longer needs the uniform bound $M(q)$: within a factor $(0.75/0.866)^b$ of the modulus-6 peak there are, for fixed $d$, only $q = 4$ and the finite list $\{2p : p \leq 2d\}$.

Figure 2 shows the measured spectrum for the first 24 primes $\geq 5$; the vanishing predicted by Lemma 3.1 is visible at $q = 10, 14, 22$, where the computed values are $6.0 \cdot 10^{-20}$, $9.0 \cdot 10^{-21}$ and $2.5 \cdot 10^{-22}$, i.e. numerically zero, against predicted heights $9.3 \cdot 10^{-4}$, $1.4 \cdot 10^{-4}$ and $6.0 \cdot 10^{-8}$ under equidistribution.

## 4 The modulus-six ripple to second order

Throughout this section $B$ is a set of $b$ odd primes with $3 \notin B$. Put

$$S_2 = \sum_{a \in B} a^2, \quad S_4 = \sum_{a \in B} a^4, \quad \Delta = \sum_{a \in B} \tau_a a, \quad \Delta_3 = \sum_{a \in B} \tau_a a^3,$$

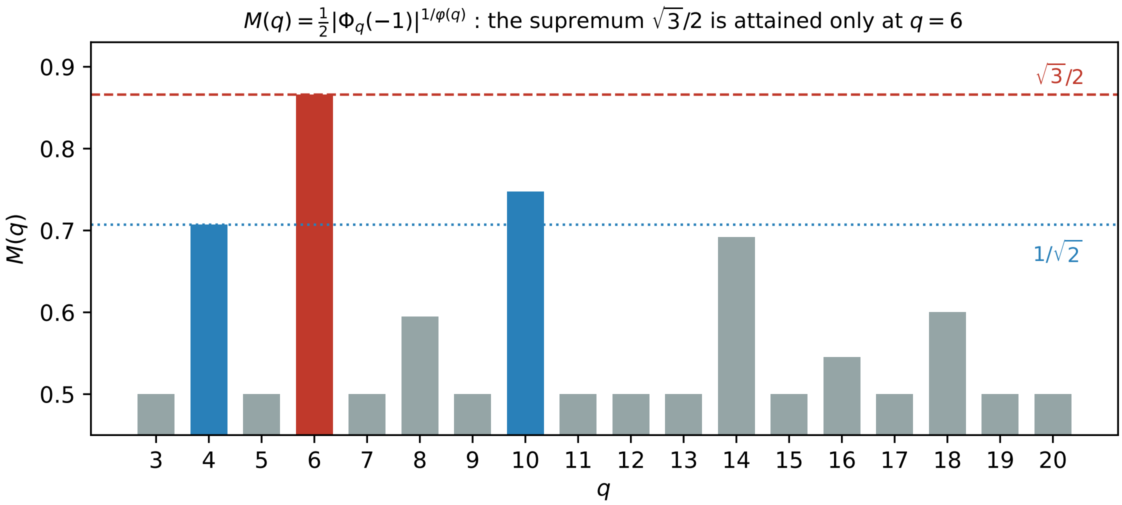


Figure 1: The per-element sub-peak height $M(q)$ (exact values, by Proposition 2.1). The supremum $\sqrt{3}/2$ is attained only at $q = 6$; $q = 10$ and $q = 4$ are the runners-up.

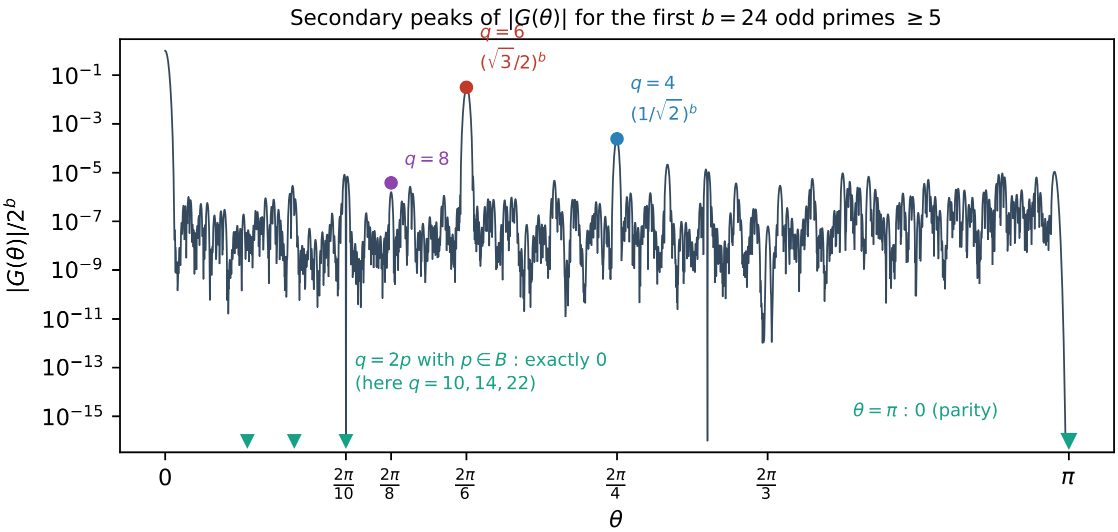


Figure 2: Measured $|G_B(\theta)|/2^b$ for the first $b = 24$ primes $\geq 5$. The peaks at $q = 2p$ with $p \in B$ are identically zero (Lemma 3.1, proved); $\theta = \pi$ vanishes by parity. Computed exactly; no asymptotics enter this figure.

where $\tau_a = +1$ if $a \equiv 1 \pmod 6$ and $\tau_a = -1$ if $a \equiv 5 \pmod 6$, and let $c_1, c_5$ be the corresponding class counts. Write $\mathrm{Main}(m)$ for the Gaussian main-arc term and $\nu = T/2 - m$.

**Lemma 4.1** (Local data at $\theta = \pi/3$)**.** *Every $a \in B$ satisfies $\cos^2(a\pi/6) = 3/4$, $\tan(a\pi/6) = \tau_a/\sqrt{3}$ and $\sec^2(a\pi/6) = 4/3$. Consequently, with $h(t) = \log|G_B(\pi/3+t)|$,*

$$h'(0) = -\frac{\Delta}{2\sqrt{3}}, \quad h''(0) = -\frac{S_2}{3} =: -V_6, \quad h'''(0) = -\frac{\Delta_3}{3\sqrt{3}}, \quad h''''(0) = -\frac{S_4}{3}.$$

*Moreover $\frac{d}{d\theta}\arg(1+e^{ia\theta}) = a/2$ exactly, so the phase of $F_B$ is linear along the arc and all curvature lives in $|G_B|$.*

(Lean: `cos_sq_pi_mul_div_six_of_one`, `cos_sq_pi_mul_div_six_of_five`, `sec_sq_pi_-mul_div_six`.)

**Proposition 4.2** (Second-order ripple formula)**.** *For $m$ in a bounded window about $T/2$,*

$$\frac{r_B(m)}{\mathrm{Main}(m)} - 1 = 2\Big(\frac{\sqrt{3}}{2}\Big)^{b+1} K_2 \cos\Big(\frac{\pi}{6}(2m - (c_1 - c_5)) + \psi(\nu)\Big) + E,$$

*where*

$$K_2 = \exp\Big[\frac{\Delta^2}{8S_2} - \frac{3}{8}\frac{S_4}{S_2^2} + \frac{\Delta\Delta_3}{4S_2^2} + \frac{5\Delta_3^2}{24S_2^3} - \frac{3S_4\Delta^2}{16S_2^3}\Big], \qquad \psi(\nu) = -\frac{\sqrt{3}}{2}\,\nu\,\frac{\Delta + \Delta_3/S_2}{S_2}.$$

*Derivation.* Laplace's method at the saddle $t^* = (L + i\nu)/V_6$ with $L = h'(0)$: the exponent contributes $(L+i\nu)^2/(2V_6) + C_3(L+i\nu)^3/(6V_6^3) + \cdots$, the Gaussian-fluctuation factor $-\frac{1}{2}\log(-h''(t^*))$ contributes $C_3 t^*/(2V_6) + C_4(t^*)^2/(4V_6) + \cdots$, and the standard constant is $5C_3^2/(24V_6^3) + C_4/(8V_6^2)$. Substituting $C_3 = -\Delta_3/(3\sqrt{3})$, $C_4 = -S_4/3$ and $V_6 = S_2/3$ from Lemma 4.1, and separating real from imaginary parts, gives $K_2$ and $\psi$. The width factor $\sqrt{V_0/V_6} = \sqrt{3}/2$ accounts for the exponent $b+1$. □

*Remark* 4.3 (The two branches)*.* The term $\Delta^2/(8S_2)$ is positive and driven by the prime race between the classes $\pm 1 \bmod 6$; the term $-\frac{3}{8}S_4/S_2^2 \asymp -9/(8b)$ is negative and class-independent. Their competition produces the two branches seen in the data. In the measured range $c_1 - c_5 \in \{-1, -3\}$, and the branch is decided by that value: $c_1 - c_5 = -3$ forces $|\Delta| \ge 135$ and $K_2 > 1$, while $c_1 - c_5 = -1$ keeps $|\Delta| \le 47$ and $K_2 < 1$.

*Remark* 4.4 (Error policy)*.* The omitted contributions are fifth- and sixth-cumulant terms and quartic crosses in $\Delta$, of order $O(|\Delta_5|/S_2^{5/2} + S_6/S_2^3 + S_4\Delta^4/S_2^4 + |\Delta_3|^3/S_2^{9/2})$ per unit of $K_2 - 1$. We state the remainder as an explicit numerical bound on the verified range together with $o(1)$ asymptotics, rather than adding further terms, which carry no structural information.

## 5 The main arc and the central density

The main arc contributes the smooth part of $r_B$. The leading term is the local central limit theorem; what matters for the companion paper's absolute formula is the *first correction* to it.

**Proposition 5.1** (Central density with first correction)**.** *Let $B$ be a set of $b$ odd integers, $V_0 = S_2/4$, and let $m$ be at the centre $T/2$. Then*

$$r_B(m) = \frac{2^b}{\sqrt{2\pi V_0}}\Big(1 - \frac{S_4}{4S_2^2} + O((S_4/S_2^2)^2)\Big).$$

*Derivation.* Write $\log G_B(\theta) = \sum_{a\in B} \log\cos(a\theta/2) = -\frac{V_0}{2}\theta^2 - \frac{S_4}{192}\theta^4 + O(S_6\theta^6)$, using $\log\cos u = -u^2/2 - u^4/12 + O(u^6)$ and $u = a\theta/2$. Since $G_B$ is even (Lean: `G_even`), all odd-order terms vanish identically, so the first correction is the quartic one. Integrating $e^{-V_0\theta^2/2}(1 - \frac{S_4}{192}\theta^4)$ over the main arc and using $\int \theta^4 e^{-V_0\theta^2/2} = 3\sqrt{2\pi}V_0^{-5/2}$ gives the stated factor $1 - \frac{S_4}{192} \cdot 3V_0^{-2} = 1 - \frac{S_4}{4S_2^2}$. □

*Remark* 5.2 (The absolute formula of the companion paper). The companion paper predicts the number of strict local minima by $\mathrm{lm} \approx \Gamma \cdot 2^b/\sqrt{2\pi V_0}$, i.e. by replacing the ground-state count by its Gaussian estimate. The measured ratio of the two sides sits systematically below 1 — about 0.97 — with no explanation offered there. Proposition 5.1 identifies the deficit as $S_4/(4S_2^2)$, which at $k = 20$ equals 0.0246. Section 6.4 verifies this. We warn against one confusion: the value 0.972 appearing here (the ratio lm/pred at $k = 20$) is *not* the 0.972 in the companion paper's list of $Q(k)$, which is $Q(18)$. Those are different quantities and the numerical coincidence is accidental: $Q$ is formed with the *measured* ground-state count, so it carries no Edgeworth correction, and it oscillates about 1 rather than sitting below it.

# 6 Numerical verification

All computations below are exact integer dynamic programming for $r_B$, followed by a discrete Fourier extraction; logs are included in the repository.

## 6.1 The amplitude correction

Table 1 compares the measured amplitude ratio against Proposition 4.2 with and without the two second-order terms.

| $\Delta_3^2$ coefficient | $S_4\Delta^2$ term | branch A mean | branch B mean |
|---|---|---|---|
| 45/24 | no | 0.00061 | 0.00736 |
| 45/24 | yes | 0.00060 | 0.00400 |
| 5/24 | no | 0.00073 | 0.00411 |
| 5/24 | yes | 0.00072 | **0.00073** |

Table 1: Mean of $|\text{measured}/K_2 - 1|$ over $k = 18, \ldots, 40$. Both corrections are needed: either one alone leaves a residual of about 0.4% on branch B.

On branch B the individual values of measured$/K_2$ are 0.9991 ($k = 32$), 0.9990 ($k = 34$) and 0.9997 ($k = 40$), against 0.9904, 0.9923 and 0.9952 without the corrections. The largest residual over the whole range is 0.0034, attained at $k = 18$. The omitted contributions are dominated by the class-independent tail $O((S_4/S_2^2)^2)$, and $S_4/S_2^2$ decreases monotonically over the range, from 0.1111 at $k = 18$ to 0.0507 at $k = 40$; hence the residual is largest at the smallest $k$. That the $\Delta$-dependent quantity $\Delta_3^2/S_2^3$ is in fact largest at $k = 32$ ($2.70 \cdot 10^{-3}$, against $1.04 \cdot 10^{-3}$ at $k = 18$) without producing an excess residual there is corroborating evidence that the $\Delta$-dependence is already absorbed into $K_2$.

## 6.2 The phase drift

Fitting the measured phase slope to $\alpha\Delta/S_2 + \beta(c_1 - c_5)/S_2 + \gamma/S_2$ over $k = 18, \ldots, 40$, only $\alpha$ is significant, at every window width tested: $t(\alpha)$ ranges from +4.26 to +5.16, while $|t(\beta)|$ and $|t(\gamma)|$ never exceed 1.6. The one-parameter model measured $= c \cdot \psi'$ gives $c = 0.884, 0.888, 0.904$ and 0.970 at half-widths 24, 48, 96 and 192, with $R^2$ from 0.978 to

0.986; restricting to $|\Delta| \geq 17$ raises $R^2$ to 0.993. We conclude that $\psi$ as stated is correct, with the deficit in $c$ a finite-window effect.

*Remark* 6.1 (Extraction). The amplitude is extracted from a narrow window of fixed half-width 24, i.e. eight full periods of the modulus-6 ripple, which over $k = 18, \dots, 40$ is between 0.08 and 0.30 standard deviations; both requirements — enough periods, and narrower than $\sigma$ — hold throughout. Wide windows corrupt the Gaussian reference. As a check, at $k = 18$ the ratio measured/$K_2$ is 0.9988, 0.9974, 1.0034 and 1.0065 at half-widths $12, 16, 24$ and 32, so the window choice moves the residual by less than 1%. The phase slope requires *wide* windows, half-width 96 to 192; at half-width 384 the normalisation breaks down in the Gaussian tails. For $|\Delta| \leq 3$ the true slope is at the level of extraction noise in narrow windows.

### 6.3 The surviving spectrum

Table 2 verifies Corollary 3.2 by varying the truncation level $d$: a peak at $q = 2p$ is dark exactly when $p$ is still present in $B_d$, and lights up as soon as the truncation removes $p$. All 50 predictions tested were correct, and $q = 6$ was the tallest surviving peak in every case.

| $d$ | removed primes | $q = 6$ | $q = 10$ | $q = 14$ | tallest |
|---|---|---|---|---|---|
| 1 | — | 0 | 0 | 0 | $q = 4$ |
| 2 | 3 | $3.66 \cdot 10^{-2}$ | 0 | 0 | $q = 6$ |
| 3 | $3, 5$ | $4.22 \cdot 10^{-2}$ | $1.03 \cdot 10^{-3}$ | 0 | $q = 6$ |
| 4 | $3, 5, 7$ | $4.88 \cdot 10^{-2}$ | $1.75 \cdot 10^{-3}$ | $4.33 \cdot 10^{-4}$ | $q = 6$ |

Table 2: $|G_{B_d}(2\pi/q)|/2^b$ for $A$ the first 24 odd primes. Entries equal to 0 are exact zeros.

Finally, at $q = 4$ and $q = 6$ the measured heights agree with $(1/\sqrt{2})^b$ and $(\sqrt{3}/2)^b$ to six decimal places for every $d$, confirming that no equidistribution hypothesis enters items (i) and (ii) of Corollary 3.2. For the remaining moduli the ratio to $M(q)^b$ lies between 0.14 and 2.3: the equidistribution approximation is *not* accurate at these sizes, which is precisely why the exact identities matter. This is to be expected, since the error in the prime number theorem in arithmetic progressions decreases only logarithmically, so at $b = 21, \dots, 31$ the $o(1)$ in items (iv)–(v) of Corollary 3.2 has not yet become small.

### 6.4 The central density

We test Proposition 5.1 using only ground-state counts. Write $\lambda(B) = \deg(B)\sqrt{2\pi V_0}/2^b - 1$ for the measured relative deficit, where $\deg(B)$ is the number of subsets attaining the minimum of $|\sum S - \lfloor T/2 \rfloor|$. Proposition 5.1 predicts $\lambda \approx -S_4/(4S_2^2)$.

| $k$ | deg | measured $\lambda$ | predicted | difference |
|---|---|---|---|---|
| 16 | 394 | $-0.024764$ | $-0.031338$ | $+0.006573$ |
| 18 | 1290 | $-0.025573$ | $-0.027754$ | $+0.002181$ |
| 20 | 4344 | $-0.024036$ | $-0.024629$ | $+0.000594$ |
| 24 | 51068 | $-0.021467$ | $-0.020830$ | $-0.000637$ |
| 28 | 634568 | $-0.017663$ | $-0.017574$ | $-0.000090$ |

Table 3: Central-density deficit for the first $k$ odd primes. The agreement is within 0.5% for $k \geq 18$ and improves to 0.01% at $k = 28$.

Over an ensemble of 100 random odd sequences per size (drawn from the same range as the primes), the mean absolute difference is 0.0171 at $k = 16$, 0.0079 at $k = 18$ and 0.0038 at $k = 20$; the ensemble is noisier than the primes because deg is a small integer subject to arithmetic fluctuation.

The consequence for Remark 5.2 is direct. Writing $\rho = \mathrm{lm}/(\Gamma \cdot 2^b/\sqrt{2\pi V_0})$ for the ratio in the absolute formula, the ensemble mean of $\rho$ is 0.9462, 0.9621, 0.9672 and 0.9716 at $k = 12, 16, 18, 20$, while the mean of $\rho/(1 - S_4/(4S_2^2))$ is 0.9812, 0.9888, 0.9912 and 0.9933. The correction removes about three quarters of the deficit, and the share it removes grows with $k$. We therefore record the systematic part of the 0.97 as explained; the residual 0.7% at $k = 20$ is a separate and smaller effect, still decreasing.

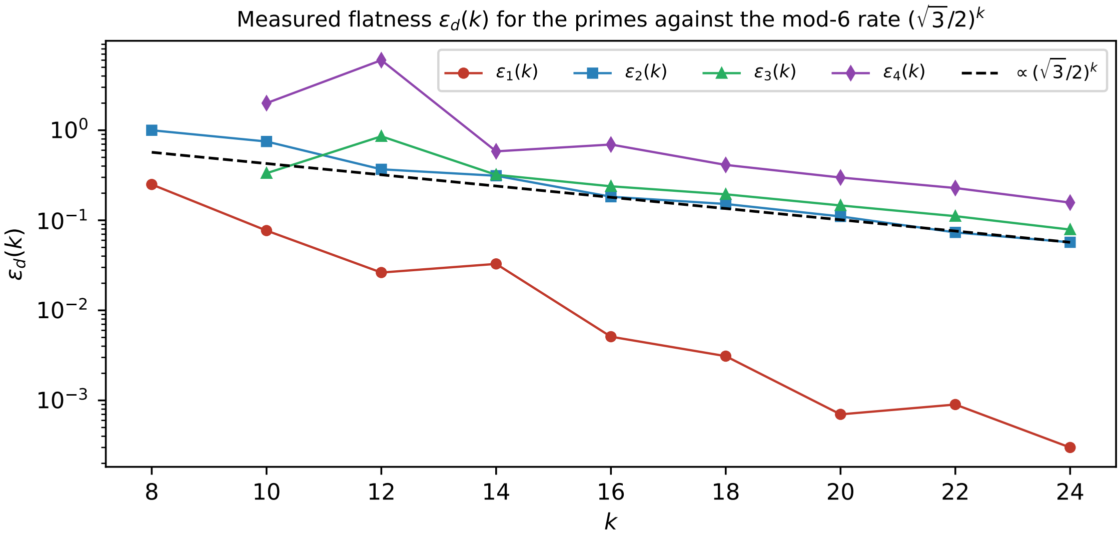


Figure 3: Measured flatness $\varepsilon_d(k)$ for the primes against the modulus-6 rate. The curve for $d = 1$ lies two orders of magnitude lower because $3 \in B_1$ and the modulus-6 peak is extinguished (Lemma 3.1). Measured by exact dynamic programming; the reference line is not a fit.

## 7 A free Gaussian envelope at every peak

The following is elementary but useful: every sub-peak carries an envelope as sharp as the main arc, with no arithmetic input at all.

**Proposition 7.1** (Curvature bound)**.** *Let $B$ be any finite set of positive reals and $V_0 = \frac{1}{4}\sum_{a\in B} a^2$. On any interval where $G_B$ has no zero,*

$$\frac{d^2}{d\theta^2} \log|G_B(\theta)| = -\sum_{a\in B} \frac{a^2}{4} \sec^2\!\Big(\frac{a\theta}{2}\Big) \;\le\; -V_0,$$

*because $\sec^2 \geq 1$. Consequently $\log|G_B|$ is concave there with curvature at least $V_0$, and if $\theta^*$ is an interior local maximum then*

$$|G_B(\theta)| \;\le\; |G_B(\theta^*)|\, e^{-V_0(\theta-\theta^*)^2/2}$$

*throughout that interval.*

(Lean: `one_le_one_div_cos_sq`, `term_curvature_ge` and `curvature_ge_V0` formalise the inequality $V_0 \le \sum(a^2/4)\sec^2(a\theta/2)$, and `deriv_le_of_second_deriv_nonpos` together with `le_of_critical_of_second_deriv_nonpos` formalise the analytic step: if $g'' \le 0$ and $g'(t) = 0$ then $t$ is a maximum of $g$. Substituting $g = f + c(\cdot - t)^2/2$ is then mechanical.)

Two caveats matter in practice, and both are visible numerically.

*Remark* 7.2 (The centre is not $2\pi/q$). The envelope must be anchored at the true local maximum, which is displaced from $2\pi/q$ by $-h'(0)/h''(0)$ with $h = \log|G_B|$; by Lemma 4.1 this is $\asymp \Delta/S_2$ at $q = 6$. For the first 24 odd primes truncated at $d = 2$ the measured displacements are $+1.316 \cdot 10^{-5}$ ($q = 6$), $+3.795 \cdot 10^{-4}$ ($q = 4$) and $-2.097 \cdot 10^{-3}$ ($q = 8$), against predicted $+1.316 \cdot 10^{-5}$, $+3.800 \cdot 10^{-4}$ and $-2.021 \cdot 10^{-3}$. Anchoring at $2\pi/q$ instead makes the envelope fail, by $7.6 \cdot 10^{-6}$ at $q = 6$ and $4.7 \cdot 10^{-3}$ at $q = 4$ in the same data.

*Remark* 7.3 (Validity radius). The bound holds only inside the zero-free interval, whose half-width is $\approx \pi/(2\max B)$. Measured in units of the envelope scale $V_0^{-1/2} = 2/\sqrt{S_2}$, this is

$$\frac{\pi}{2\max B} \cdot \frac{\sqrt{S_2}}{2} \;\approx\; \frac{\pi}{4}\sqrt{k/3} \;\approx\; 0.45\sqrt{k},$$

so the envelope covers a growing number of standard deviations: 1.72 at $k = 16$, 2.08 at $k = 24$, 2.71 at $k = 40$ and 2.92 at $k = 48$, against the prediction $1.80, 2.21, 2.85, 3.12$.

# 8 Effective constants for the surviving peaks

Items (iv) and (v) of Corollary 3.2 are the only ones that use equidistribution, and only for a finite list of moduli. We make them effective.

Write $n_r = \#\{p \in B_d : p \equiv r \pmod q\}$ for $\gcd(r, q) = 1$, and $e_r = n_r - b/\varphi(q)$. Since $\log|G_{B_d}(2\pi j/q)| = \sum_r n_r \log|\cos(\pi r j/q)|$ and $\sum_r e_r = O(1)$,

$$|G_{B_d}(2\pi j/q)| \;\le\; M(q)^b \exp\Big(L_q \sum_r |e_r|\Big), \qquad L_q = \max_{\gcd(r,q)=1} \big|\log|\cos(\pi r/q)|\big|.$$

Bennett, Martin, O'Bryant and Rechnitzer [BMOR18] supply two explicit inputs which between them cover all $x$: their Theorem 1.3 gives $|\pi(x; q, a) - \mathrm{Li}(x)/\varphi(q)| < c_\pi(q)\, x/(\log x)^2$ for $x \ge x_\pi(q)$, with $c_\pi(q) \le 1/840$ and $x_\pi(q) \le 8 \cdot 10^9$ for $3 \le q \le 10^4$; and their Theorem 1.9 gives $\max_{y \le x} |\pi(y; q, a) - \mathrm{Li}(y)/\varphi(q)| \le 2.734\sqrt{x}/\log x$ for $x \le x_2(q)$, where $x_2(q) = 10^{13}$ for $5 < q \le 100$. The ranges overlap, so there is no gap.

**Proposition 8.1** (Effective form of Corollary 3.2(iv)–(v))**.** *For each $\eta > 0$ and each modulus $q$ in the finite list there is an explicit $k_0(q, \eta)$ such that $|G_{B_d}(2\pi j/q)| \le (M(q) + \eta)^b$ for all $k \ge k_0(q, \eta)$. With the constants of [BMOR18] one may take*

| $q$ | 4 | 6 | 8 | 10 | 14 | 18 |
|---|---|---|---|---|---|---|
| $k_0(q, 0.10)$ | 108 | 34 | 1694 | 856 | 3547 | 4411 |
| $k_0(q, 0.05)$ | 343 | 104 | 5596 | 2839 | 11914 | 14723 |
| $k_0(q, 0.01)$ | 5984 | 1737 | 103960 | 52658 | 226200 | 278350 |

*Remark* 8.2. The $\sqrt{x}$-shaped bound of [BMOR18, Thm. 1.9] is what makes these thresholds small. Using only [BMOR18, Thm. 1.3] one is forced to $x \ge x_\pi(q) \approx 10^9$, i.e. $k_0 \approx 4 \cdot 10^8$; the two theorems must be combined. It is worth recording that using only the first of the two costs a factor of $10^4$ in the threshold.

Regarding the range of validity we are explicit in three steps. *(a)* For $k \le 44$ the peak heights are certified by direct computation: $|G_{B_d}(2\pi j/q)|$ is a product of $b$ cosines and costs $O(k)$ arithmetic operations, so it is simply evaluated (Section 6). *(b)* For $k \ge k_0(q, \eta)$ the bound is a theorem, by Proposition 8.1. *(c)* The gap between the two is not needed for any asymptotic statement, and in any case it can be closed by computation: for each of the finitely many $q$ involved, evaluating $|G_{B_d}(2\pi j/q)|$ for all $k$ up to any desired bound is an $O(k)$ computation per point. Note also that $q = 4$ and $q = 6$ appear in the table only for comparison: for those two moduli Corollary 3.2(i)–(ii) are exact identities and no equidistribution input is needed at all.

# 9 The deep minor arcs

Most of this section is now proved: the dissection (Lemma 9.1), the counting of bad points (Lemma 13.1), the twisted sums (Lemma 11.9), the annulus profile with its closed form and its consequence (Lemmas 11.11, 11.13, 11.4). What is not written out to referee standard is the substitution of the quoted exponential-sum bound in step 2 and the excision of step 4; we flag those explicitly where they occur, and Section 11 keeps the ledger.

## 9.1 The product is a superposition of prime exponential sums

Expanding $\log|2\cos\pi u| = \sum_{n\geq 1}(-1)^{n+1}\cos(2\pi nu)/n$ termwise,

$$\log|G_B(\theta)| = -b\log 2 + \sum_{n\geq 1}\frac{(-1)^{n+1}}{n}\operatorname{Re}\sum_{a\in B} e\Big(\frac{na\theta}{2\pi}\Big). \tag{1}$$

The inner sum is a classical exponential sum over primes. Thus the object we must bound on the deep minor arcs is not exotic: it is a weighted superposition of the sums $\sum_{p\leq x} e(\alpha p)$ that Vinogradov's method was built for, with $\alpha = n\theta/2\pi$. The only delicacy is that the coefficients $1/n$ decay slowly, and that the series diverges at the zeros of cos, so a truncation is needed near those points.

## 9.2 How the circle is divided, and why nothing falls between

Before the argument we fix the dissection, because the choice of the truncation parameter $N_0$ decides whether it is exhaustive. Let $A \geq 2$ be the exponent in the Siegel–Walfisz theorem, put $A' = A + 1$, and set

$$\delta_0 = \frac{1}{\log b}, \qquad N_0 = (\log N)^{A'}, \qquad Q_2 = (\log N)^{A+A'}.$$

| Tier | range of $q$ | tool | located in |
|---|---|---|---|
| exact peaks | $q \in \{4,6\} \cup \{2p : p \in B\}$ | cyclotomic identities | §3 |
| $T_2$ | remaining $q \leq Q_2$ | PNT in AP + envelope | §8 |
| $T_3$ | deep minor, see (ii) below | Vaughan + Selberg + Abel | this section |

In tier $T_2$ the equidistribution of $B$ in the residue classes mod $q$ is what produces the constant $M(q)$. Asymptotically this is the Siegel–Walfisz theorem, which is what fixes $A$; effectively, for the $q$ that occur at any accessible $k$, it is the explicit bounds of [BMOR18] used in Proposition 8.1. Only the ineffectivity of the Siegel zero prevents the two from being the same statement, and that is the sole reason Section 8 carries a threshold rather than a constant.

**Lemma 9.1** (Covering completeness)**.** *Every $\theta$ satisfies at least one of*

*(i) $|\theta - 2\pi j/q| \leq 1/(q\tau)$ for some $q \leq Q_2$ and $(j,q) = 1$;*

*(ii) for every $n \leq N_0$ the reduced denominator $q_n$ of $n\theta/2\pi$ exceeds $(\log N)^A$.*

*Proof.* Suppose (ii) fails: some $n \le N_0$ has $q_n \le (\log N)^A$, so $|n\theta/2\pi - j'/q_n| \le 1/(q_n\tau)$ for some $j'$. Dividing by $n$ exhibits $\theta/2\pi$ within $1/(nq_n\tau)$ of $j'/(nq_n)$, and $nq_n \le N_0(\log N)^A = Q_2$. Reducing the fraction only decreases the denominator, which gives (i). □

*Remark* 9.2. With the earlier choice $N_0 = b^{1/4}$ the same computation gives $nq_n \le b^{1/4}(\log N)^A$, and the band $(\log N)^A < q \le b^{1/4}(\log N)^A$ then belongs to no tier: too large for the explicit bounds, too small to be deep minor. Replacing $b^{1/4}$ by a fixed power of $\log N$ closes it, and costs nothing — see Remark 9.7, where the change in fact improves every measured quantity that matters.

### 9.3 The intended argument

Fix $\theta$ of type (ii); the $\theta$ of type (i) are covered by Corollary 3.2 and Proposition 8.1. Split $B = B_{\text{good}} \cup B_{\text{bad}}$, where $p$ is bad when $\|p\theta/2\pi - \frac{1}{2}\| < \delta_0$.

1. *Harmonics stay minor.* This is exactly condition (ii): for every $n \le N_0$ the reduced denominator of $n\theta/2\pi$ exceeds $(\log N)^A$, so every harmonic appearing below lies on a minor arc. This is the only place where the value of $N_0$ enters the dissection, and it is why $N_0$ must be a power of $\log N$ rather than of $b$ (Remark 9.2).

2. *The exponential sums.* Take the Dirichlet parameter $\tau = N(\log N)^{-A}$, so that every denominator arising above satisfies $q_n \le \tau$ automatically; combined with step 1,
$$(\log N)^A \;<\; q_n \;\le\; N(\log N)^{-A},$$
which is precisely the range on which Vaughan's identity is effective. No other range occurs: the tiers below $(\log N)^A$ are not lost but are handled in §3 and §8, where the cyclotomic constants $M(q)$ give the bound directly and no exponential sum is needed at all. On this range Vaughan's identity gives
$$S_B(\alpha) = \sum_{p\in B} e(p\alpha) \ll (\log N)^3\big(Nq_n^{-1/2} + N^{4/5} + (Nq_n)^{1/2}\big),$$
valid whenever $1 \le q_n \le N$, $(a, q_n) = 1$ and $|q_n\alpha - a| \le q_n^{-1}$; this is Lemma 3.1 of Kumchev's survey [Ku05], where it is attributed to Vaughan as the sharpest form of Vinogradov's 1937 estimate. Our Dirichlet parameter gives $|q_n\alpha - a| \le 1/\tau \le 1/q_n$, so the hypotheses hold. Hence $S_B \ll N(\log N)^{3-A/2}$, which, since $\pi(N) \asymp N/\log N$, is $o(\pi(N))$ as soon as $A > 8$; $A$ is ours to choose and Siegel–Walfisz holds for every fixed $A$, so we fix $A = 11$ once and for all. A log-free version valid on a wider range of $q$ is available in [Sr25]. Summing the head, $\sum_{n\le N_0} |S_B(n\theta)|/n \le (\log N_0)\max_n |S_B(n\theta)|$, gives $o(b)$; note that $\log N_0 = A' \log\log N$, so enlarging $N_0$ from $b^{1/4}$ to $(\log N)^{A'}$ costs the head only a bounded factor.

3. *Bad points.* Each bad $p$ contributes at most $\log 2$ to (1), so it suffices that $|B_{\text{bad}}| = o(b)$. This needs no input beyond step 2. Lemma 13.1 of the appendix — the one-interval form of the Erdős–Turán inequality, proved there from the Fejér kernel so that no external reference is incurred — applied to the interval $\|x - \frac{1}{2}\| < \delta_0$ with degree $N_0$, gives
$$|B_{\text{bad}}| \;\le\; 4b\Big(\delta_0 + \frac{1}{N_0}\Big) + 4\sum_{n\le N_0} c_n\,|S_B(n\theta)|, \qquad c_n = \min\Big(2\delta_0 + \tfrac{2}{N_0}, \tfrac{1}{\pi n}\Big),$$
and the sum on the right is the *same* head already shown to be $o(b)$ in step 2. Since $\delta_0 \to 0$ and $N_0 \to \infty$, the whole expression is $o(b)$.

We stress the point, because it removes a dependency rather than adding one: no external discrepancy theorem for $\{p\theta\}$ is required. The literature on $\alpha p$ modulo one — Vinogradov, Vaughan, Harman, Matomäki — proves *small-value* statements, that some prime $p \le N$ makes $\|\alpha p - \beta\|$ small; that is a statement about the extreme of the sequence and does not by itself bound its discrepancy. The route above uses only the exponential sums we already have.

4. *Good points.* For $p$ good the series (1) is truncated at $N_0$; since $\|p\theta/2\pi - \frac{1}{2}\| \ge \delta_0$, summation by parts against the Dirichlet kernel bounds the tail by $C/(N_0\delta_0) = C \log b\, (\log N)^{-A'} = o(1)$. Here the new choice of $N_0$ is a large gain and not merely a repair: with $N_0 = b^{1/4}$ the same bound was $C \log b \cdot b^{-1/4}$, which stays above 1 until $b \approx 10^6$, whereas $(\log N)^{A'}$ makes it small immediately (Remark 9.7).

Together these give $\log |G_B(\theta)| \le -(b - o(b)) \log 2 + o(b)$, that is $|G_B(\theta)| \le (1/2 + o(1))^b$ on the deep minor arcs.

*Remark* 9.3 (What this settles about the technique needed). Only a one-sided bound together with an exceptional set of size $o(b)$ is required. The precise asymptotic machinery of [DRZZ25] — their strange and pseudo-differentiable functions — is therefore not needed here, and we cite them for context only. The question was nevertheless the right one: the non-smoothness of $\log |2\cos|$ is exactly where the elementary excision of step 4 does its work.

*Remark* 9.4 (An alternative route). One can instead avoid (1) altogether and argue through the full discrepancy $D_B(\theta)$ of $\{p\theta/2\pi\}_{p\in B}$: bound it by Erdős–Turán from the same exponential sums, then apply Koksma's inequality to the truncated function $\max(\log|\cos \pi x|, \log \delta)$. The two routes give the same conclusion. We prefer the one above for two reasons: (1) is an exact identity, and step 3 needs the Erdős–Turán machinery only for a single interval, where it is elementary, rather than uniformly over all intervals.

*Remark* 9.5 (Where the effective constants are, and are not). Effectivity in this paper is confined to three places: the peak heights at $q = 4, 6$, which are exact identities (Corollary 3.2(i)–(ii)); the remaining finite list of moduli, effective with $k_0 \le 1.5 \cdot 10^4$ (Proposition 8.1); and the Gaussian envelopes, which need no arithmetic input at all (Proposition 7.1). This section is asymptotic; see Remark 9.6.

*Remark* 9.6 (On explicit constants). One might hope to make step 3 effective using Helfgott's explicit minor-arc bounds [Hel12]. His Main Theorem is stated for $x \ge x_0 = 2.16 \cdot 10^{20}$, which corresponds to $k = \pi(x_0) \approx 4.7 \cdot 10^{18}$. That is far outside any range we could combine with the direct computations of Section 6, so we do not claim an effective version of this section. Effectivity in this paper is confined to Proposition 8.1 (thresholds $\approx 10^4$) and to Proposition 7.1, which needs no arithmetic input at all.

*Remark* 9.7 (Numerical check of the budget). The parameters are asymptotic and are meaningless at $b = 23$, so we tested the steps above at $b$ up to 1999, taking the worst of six sampled deep-minor $\theta$ at each size. Each evaluation is $O(bN_0)$. One substitution is forced: the argument takes $N_0 = (\log N)^{A'}$ with $A' = 12$, which at $b = 1999$ is about $2 \cdot 10^{10}$ and cannot be computed; the table therefore uses $N_0 = (\log b)^3$, which is the same qualitative object — a fixed power of a logarithm — and is the smallest such choice for which the tail is already meaningful. Increasing the exponent only improves the tail and worsens the head, both monotonically, so the comparison with $b^{1/4}$ below is not sensitive to it. Writing tail for the step-4 bound $1/(N_0\delta_0)$:

| $b$ | 99 | 299 | 999 | 1999 |
|---|---|---|---|---|
| $\lvert B_{\mathrm{bad}}\rvert/b$ (measured) | 0.485 | 0.405 | 0.324 | 0.282 |
| step-3 bound$/b$, $N_0=(\log N)^3$ | 1.079 | 0.910 | 0.614 | 0.459 |
| $b^{-1}\sum_{n\le N_0}\lvert S_B(n\theta)\rvert/n$ | 0.697 | 0.640 | 0.415 | 0.292 |
| tail, $N_0=b^{1/4}$ (old) | 1.532 | 1.425 | 1.381 | 1.267 |
| tail, $N_0=(\log N)^3$ (new) | 0.047 | 0.031 | 0.021 | 0.017 |

The bound of step 3 holds in every case. The change of $N_0$ is the decisive one: with $N_0=b^{1/4}$ the tail bound never drops below 1 in any accessible range — it would need $b\approx 10^6$ — so that step was vacuous in practice, whereas with $N_0=(\log N)^3$ it is already below 0.05 at $b=99$. The price is a larger head, since more harmonics are summed; the head is the slower of the two now, and the step-3 bound only becomes nontrivial (that is, smaller than 1) from $b\approx 300$.

None of this is close to binding. The conclusion needs only $\frac{1}{2}e^{o(1)}<\sqrt{3}/2$, i.e. the $o(1)$ to stay below $\log\sqrt{3}=0.549$, and the quantity it controls, $-b^{-1}\log|G_B(\theta)|-\log 2$, was measured at 0.123 for $b=99$ and 0.024 for $b=1999$.

*Remark* 9.8 (What the peaks actually look like)*.* As a check on the dissection we located every local maximum of $|G_B|$ on $[0,\pi]$ by direct evaluation on a grid of $6\cdot 10^6$ points, for $k=40,60,100,140$. Every one of the twelve largest lies within the envelope radius of a rational $2\pi j/q$ with $q\le 26$, and the ordering by height follows $M(q)$ as predicted, with one instructive exception: at $q=10$, where $5\in B$ forces $G_B(2\pi/10)=0$ exactly, the vanishing cuts a narrow notch and the flanking maxima sit at $b^{-1}\log|G_B|=-0.355$ at $b=99$, rising towards $\log M(10)=-0.291$ like $-\frac{3}{2}(\log b)/b$, which is the expected width of the notch. Excluding the central peak and a disc of the envelope radius about each rational with $q\le 60$, the largest remaining value of $b^{-1}\log|G_B|$ is $-0.305$ at $b=99$ and $-0.305$ at $b=139$, against $\log(\sqrt{3}/2)=-0.144$: a margin of 0.16 per element, stable in $k$.

## 10 The main theorem

Nothing new is proved in this section; it is the inventory that assembles the previous ones.

**Theorem 10.1** (Asymptotic flatness, conditional)**.** Assume *(a)* the deep-minor bound of Section 9*, i.e. that* $|G_B(\theta)|\le(1/2+o(1))^b$ *for* $\theta$ *of type (ii) of Lemma 9.1 — that section establishes it at the level of a sketch —* and *(b)* the bound $\|\Phi_q\|_1\le 3^{\varphi(q)/2}$ on the range of Problem 11.8*, verified by exact integer arithmetic for* $q\le 1200$*. Fix* $d\ge 2$ *and let* $n=\lfloor T/2\rfloor$*. With* $B=B_d(k)$*,* $b=|B|$*, as* $k\to\infty$*,*

$$\varepsilon_d(k)=4\Big(\frac{\sqrt{3}}{2}\Big)^{b+1}K_2(k)\,\kappa_d(k)\,\big(1+o(1)\big)+O\Big(\frac{w_d^2}{S_2}\Big),$$

*where* $K_2$ *is the amplitude correction of Proposition 4.2,* $\kappa_d\in[0,1]$ *is the explicit trigonometric window factor — half the range of* $\cos((\pi/6)(2m-(c_1-c_5))+\psi)$ *over* $m\in\mathrm{Win}_d$ *— and* $w_d$ *is the window width. In particular* $\varepsilon_d(k)\to 0$ *at the geometric rate* $\sqrt{3}/2$ *per element.*

*Remark* 10.2. The hypotheses are not decorative: they are the two statements of this paper that are not proved, and we prefer to carry them explicitly rather than bury them. Hypothesis (a) disappears once the programme of Section 9 is written out in full; hypothesis (b) is a bounded amount of work on the envelope and is supported by the measurements of Remark 9.8.

**Corollary 10.3** (conditional, as above)**.** *Combining Theorem 10.1 with the sandwich theorem of the companion paper and the identity $W_D = \Gamma + (2D+1)2^{-k}$,*

$$\frac{\mathrm{lm}_P(n)}{\deg_P(n)} \longrightarrow \Gamma(P) = 5.34920\ldots$$

*for the odd primes.*

*Outline.* The bookkeeping is as follows. *(1)* By Proposition 5.1, $r_B(m) = \mathrm{Main}(m)(1+E_0(m))$ with the correction $1 - S_4/(4S_2^2)$; over the bounded window the smooth ratio varies by $O(w_d^2/S_2)$. *(2)* By Proposition 4.2, the two exact peaks contribute $2(\sqrt{3}/2)^{b+1}K_2\cos(\cdot+\psi)$ and $2(1/\sqrt{2})^{b+1}K_4\cos(\cdot)$; the second is smaller by a factor $(0.8165)^b$ and is absorbed into $1+o(1)$. *(3)* By Lemma 3.1 every peak at $q = 2p$ with $p \in B$ vanishes, and by Corollary 3.2 and Proposition 8.1 the surviving finite list has heights at most $(m_d+\eta)^b$ with $m_d \le 5^{1/4}/2$. *(4)* By Proposition 7.1 each peak carries a Gaussian envelope of curvature $V_0$ centred at its true maximum, with reach $\Theta(\sqrt{k})$ standard deviations, so the arcs tile. *(5)* Off all envelopes, Section 9 gives $|G_B| \le (1/2+o(1))^b$. *(6)* Integrating over each region and dividing by Main gives the inventory

| region | bound, relative to Main | measured, $k=32$ |
|---|---|---|
| central | 1 | 0.985 |
| $q=6$ | $C(\sqrt{3}/2)^b$ | $1.03\cdot 10^{-2}$ |
| $q=4$ | $C(1/\sqrt{2})^b$ | $1.48\cdot 10^{-5}$ |
| other $q \le Q_2$ | $CQ_2\,(m_d+o(1))^b$ | $4.06\cdot 10^{-7}$ |
| $q=2p,\ p\in B$ | $CQ_2\,(M(2p)+o(1))^b b^{-3/2}$ | — |
| deep minor | $CQ_2\,(m_d+o(1))^b + C(1/2+o(1))^b$ | $3.66\cdot 10^{-6}$ |

with $m_d \le M(18) = 0.6005$ for $d=2$. The row for $q=2p$ is there because the vanishing of Corollary 3.2(iii) is pointwise: it contributes no mass at the rational itself but the notch is narrow and the flanking peaks are not removed (Remark 11.16). The last row deserves a word. It is tempting to bound the deep-minor region by its pointwise bound $(1/2+o(1))^b$ times the measure of the region, but that is wrong by several orders of magnitude — at $k=32$ it would give $1.2\cdot 10^{-7}$ against a measured $3.7\cdot 10^{-6}$ — because the integral there is dominated not by the deepest points but by the many shallow peaks the region also contains, each of which carries a width. Counting those peaks with their widths, at height $m_d^b$, reproduces the measurement to within a bounded factor. The conclusion is unaffected: what matters is that every row after the second is $o((\sqrt{3}/2)^b)$, and the measured ratios of rows 3–6 to row 2 fall from 1.3% at $k=24$ to 0.04% at $k=32$. Taking the maximum and minimum over the bounded window then gives the displayed formula. □

## 11 Honest scope, and what remains

We set out plainly which parts of this paper are proved and which are not.

- *Proved on paper*: Proposition 2.1 and Lemma 2.2 (classical), Theorem 2.3, Lemma 3.1, Corollary 3.2(i)–(iii) and (vi), Proposition 7.1, Lemma 9.1, Lemma 13.1, Lemma 12.2, Lemma 11.9, Lemma 11.11, Lemma 11.13, Lemma 11.4, Lemma 11.7, Theorem 14.1.
- *Derivations verified numerically to stated precision*: Proposition 4.2 ($K_2$ and $\psi$, to 0.1% over $k=18,\ldots,40$) and Proposition 5.1 (the Edgeworth term).
- *Effective*: Proposition 8.1 ($k_0 \le 1.5\cdot 10^4$) and Corollary 3.2(i)–(ii), which are exact.

- *Sketch level, to be written to referee standard*: Section 9 (the Vinogradov input of step 2 and the excision of step 4), Proposition 14.3, and the $o(1)$ bookkeeping in item (6) of the outline above. Theorem 10.1 and Corollary 10.3 inherit this status.
- *A gap we have found and not closed.* Lemma 9.1 sorts every $\theta$ by *denominator*, but the two mechanisms it hands $\theta$ to are matched to different *radii*: the Dirichlet neighbourhood in case (i) has radius $1/(q\tau)$, while the envelope of Proposition 7.1 is certified only out to $\approx 0.45\sqrt{k}\,\sigma$, smaller by a power of $\log N$. Nothing in this paper bounds $|G_B|$ on the annulus between the two. The measurements of Remark 9.8 say the annulus is harmless — the largest value anywhere outside the envelopes is 0.16 per element below $\log(\sqrt{3}/2)$, stably in $k$ — but that is a measurement, not a proof. The repair is Lemma 11.11, which replaces the question rather than answering it: a profile valid at every distance makes the matching of radii unnecessary. That lemma is now proved; what survives of the gap is Problem 11.8: by Lemmas 11.4 and 11.7 it is the integer inequality $\|\Phi_q\|_1 \leq 3^{\varphi(q)/2}$, verified exactly for $q \leq 1200$ and needed for all squarefree $q \geq 7$. No range of $x$ has to be examined: see Remark 11.5. We warn that the naive form of the inequality — $\Phi_q \leq 0$ — is *false*, for the reason given in Remark 11.14.
- *Formal verification*: the 99 Lean theorems cover the combinatorial core — the identities of both papers, the curvature bound, the vanishing lemma, and now Lemma 11.7 in the form `norm_poly_le_l1_on_circle`. The analytic sections are not formalised and we do not claim that they are.

The remaining work is therefore of three kinds, and none of it requires a new idea: writing Section 9 and Proposition 14.3 out in full, discharging the $o(1)$ inventory, and the usual polishing. We state the three steps as problems for the record.

**Problem 11.1** (Main arc)**.** Make the local central limit theorem for $r_B$ on the main arc effective, with an error term of the shape $C/S_2$ arising from the curvature sum of the smooth part. Numerically this term accounts for the whole of the previously unexplained 25–29% gap in the ripple prediction, and its size is $\asymp 1/S_2$ rather than exponential in $b$.

**Problem 11.2** (Minor arcs)**.** Carry out the programme of Section 9. The exponential sums are the classical ones by (1); the non-classical part is the excision near the zeros of cos. Dong, Robles, Zaharescu and Zeindler [DRZZ25] handle non-principal major arcs for prime partitions using what they call strange and pseudo-differentiable functions, and their treatment of non-smooth behaviour looks like the right tool for that excision, although their limit ($n \to \infty$ with the part set fixed) is not ours.

**Problem 11.3** (The annulus)**.** Show that

$$\log M(q) + \Phi_q(x) \;\leq\; \log\frac{\sqrt{3}}{2} \qquad \text{for } x_{\mathrm{env}}\ (\approx 1.5) \leq x \leq \frac{2\pi^2}{3\log(2M(q))}$$

and every $q \leq Q_2$. *This is now a lemma; see Lemma 11.4 below.*

**Lemma 11.4** (The annulus, resolved)**.** *For every $q$ and every $x > 0$,*

$$\log M(q) + \Phi_q(x) \;\leq\; \frac{1}{\varphi(q)}\log\|\Phi_q\|_\infty - \log 2, \qquad \|\Phi_q\|_\infty := \max_{|z|=1}|\Phi_q(z)|.$$

*Consequently* $\log M(q) + \Phi_q(x) \leq \log(\sqrt{3}/2)$ *for all $x > 0$ as soon as* $\|\Phi_q\|_\infty \leq 3^{\varphi(q)/2}$.

*Proof.* By Lemma 11.13, $\log M(q) + \Phi_q(x) = -\log 2 + \frac{1}{x}\int_0^x h_q$ where $h_q(v) = \varphi(q)^{-1}\log|\Phi_q(-e^{-iv})|$. Bound the integrand by its supremum, $h_q(v) \leq \varphi(q)^{-1}\log\|\Phi_q\|_\infty$, and average. $\square$

*Remark* 11.5 (What this replaces). This is worth stating plainly, because it is the third and last form the question took. We first asked for the reach of the envelope, then for the sign of $\Phi_q$ on a compact range of $x$, and expected to settle the latter by certified numerical integration. Once Lemma 11.13 had put $\Phi_q$ in the form of a running average, no integration was needed: a running average never exceeds the supremum of what is averaged, so *no range of $x$ has to be examined at all.* The tight case $q = 6$ is immediate from $|1 + 2\cos v| \le 3$.

The same identity explains Remark 11.14 exactly: $\Phi_q \le 0$ holds precisely when $\|\Phi_q\|_\infty = |\Phi_q(-1)|$, that is, when $|\Phi_q|$ attains its maximum on the circle at $z = -1$. For $q = 4, 6, 8, 10, 18$ it does; for $q = 12, 15, 20, 21, 24$ it does not, and those are exactly the counterexamples of Remark 11.14.

*Remark* 11.6 (Large $q$ needs nothing new). For $n \le N_0$, Hölder's formula $c_q(n) = \mu(q/g)\varphi(q)/\varphi(q/g)$ with $g = \gcd(n, q) \le N_0$ gives $|c_q(n)|/\varphi(q) \le 1/\varphi(q/g)$, and $\varphi(m) \gg m/\log\log m$ then yields $|c_q(n)|/\varphi(q) \ll N_0 \log\log q/q$. (One may not write $\varphi(q/g) \ge \varphi(q/N_0)$: $\varphi$ is not monotone.) Hence $\Phi_q \to 0$ uniformly as $q \to \infty$, and $\log M(q) + \Phi_q \le \log(\sqrt{3}/2) + o(1)$ by Theorem 2.3. So only a finite list of $q$ is ever at issue in Problem 11.8.

**Lemma 11.7** ($\ell^1$ reduction)**.** $\|\Phi_q\|_\infty \le \|\Phi_q\|_1 := \sum_i |c_i|$*, where* $\Phi_q = \sum_i c_i x^i$*.*

*Proof.* $|\Phi_q(z)| = |\sum_i c_i z^i| \le \sum_i |c_i||z|^i = \sum_i |c_i|$ for $|z| = 1$. (Formalised: `norm_poly_le_l1_on_circle`.) □

Since $\varphi(q)$ is even for $q \ge 3$, the quantity $3^{\varphi(q)/2}$ is an integer, so the criterion of Lemma 11.4 becomes a comparison of two integers:

$$\sum_i |c_i| \;\le\; 3^{\varphi(q)/2}.$$

We have verified this by exact integer arithmetic for $3 \le q \le 1200$. It holds throughout, with equality *only* at $q = 3$ and $q = 6$. Writing $\lambda(q) = \varphi(q)^{-1} \log \|\Phi_q\|_1$ for the quantity that must stay below $\log\sqrt{3} = 0.5493$:

| $Q$ | 3 | 7 | 13 | 61 | 241 | 481 | 961 |
|---|---|---|---|---|---|---|---|
| $\sup_{q \ge Q} \lambda(q)$ | 0.5493 | 0.4024 | 0.3243 | 0.1354 | 0.0526 | 0.0337 | 0.0182 |

So $Q_4 = 7$ suffices: past the two extremal moduli $q = 3, 6$ the margin is already 0.147 per element, and it grows. The $\ell^1$ norm itself barely moves — it is 7 at $q = 240$, against $3^{32}$ on the right.

Two structural remarks cut the problem down further. First, $\Phi_n(x) = \Phi_{\mathrm{rad}(n)}(x^{n/\mathrm{rad}(n)})$, so $\|\Phi_n\|_1 = \|\Phi_{\mathrm{rad}(n)}\|_1$ exactly, while $\varphi(n) \ge \varphi(\mathrm{rad}(n))$; hence $\lambda(n) \le \lambda(\mathrm{rad}(n))$ and *only squarefree $q$ need be considered.* (Verified for $3 \le n \le 300$: no exceptions.) Second, the extremal moduli are exactly the ones with the smallest $\varphi$: $q = 3, 6$ give $\|\Phi_q\|_1 = 3 = 3^{\varphi(q)/2}$, and every larger squarefree modulus has enough room in $\varphi(q)$ to absorb its $\ell^1$ norm.

**Problem 11.8** (The remaining range)**.** Prove $\lambda(q) < \log\sqrt{3}$ for all squarefree $q \ge 7$, not merely for $q \le 1200$. Equivalently, bound $\log \|\Phi_q\|_1$ by $\frac{1}{2}\varphi(q)\log 3$. Three remarks on where the difficulty is and is not.

*(a)* It is not a question about $x$: by Lemma 11.4 the whole $x$-dependence has been discharged, and by the paragraph above only squarefree $q$ matter.

*(b)* The trivial pointwise bound $h_q \le \log 2$ is tantalisingly close but not enough — 0.693 against the 0.549 wanted, a factor 1.26 — and it cannot be improved without using that the reduced residues mod $q$ are spread around the circle.

*(c)* A termwise estimate of the series for $\Phi_q$ will not do it. Using $|\widetilde{I}(mx)| \le 2/(mx)$ on the range $m > \sqrt{q}$ with $x \gg 1/q$ gives $\sum_{m>\sqrt{q}}(1/m)(2q/(m\delta)) \le 2\sqrt{q}/\delta$, which *diverges*; inserting the Ramanujan-sum decay $|c_q(m)|/\varphi(q) \ll m\log\log q/q$ still leaves $\log q \log\log q/\delta$. The loss is the oscillation in $m$, which termwise bounds throw away. The natural tool instead is the identity $\prod_{s=0}^{Q-1} 2\sin\pi(x+s/Q) = 2\sin(\pi Q x)$, which collapses each arithmetic progression in a Möbius decomposition of $\sum_{r\perp q}$ to a single $\log|2\sin|$; the obstacle there is that the terms with $\mu(d) = -1$ need a *lower* bound, and $\log|2\sin|$ is unbounded below.

**Lemma 11.9** (Twisted sums, uniformly). *Let $q \le (\log N)^A$, let $r$ be coprime to $q$, let $n \le N_0$ and let $|t| \le 1/(q\tau)$. Then*

$$\sum_{\substack{p\in B\\ p\equiv r\ (q)}} e(pnt) \;=\; \frac{1}{\varphi(q)}\sum_{p\in B} e(pnt) \;+\; O(Ne^{-c\sqrt{\log N}}(\log N)^{A+A'}),$$

*uniformly in $t$, $n$ and $r$.*

*Proof.* Put $f(u) = e(unt)$, $F(u) = \#\{p \in B : p \le u,\ p \equiv r\ (q)\}$ and $G(u) = \#\{p \in B : p \le u\}$. Abel summation gives $\sum_{p\equiv r} e(pnt) = f(N)F(N) - \int_{2d}^{N} F(u)f'(u)\,du$ and the same identity with $G/\varphi(q)$ in place of $F$. Subtracting and writing $E(u) = F(u) - G(u)/\varphi(q)$,

$$\Big|\sum_{p\equiv r} e(pnt) - \frac{1}{\varphi(q)}\sum_{p} e(pnt)\Big| \;\le\; |E(N)| + \int_{2d}^{N} |E(u)|\,|f'(u)|\,du \;\le\; E^*(1 + 2\pi N|nt|),$$

since $|f'(u)| = 2\pi|nt|$. Two inputs finish it. First, $E^* = \sup_{u\le N}|E(u)| \ll Ne^{-c\sqrt{\log u}}$ uniformly for $q \le (\log N)^A$, by the Siegel–Walfisz theorem applied to $\pi(u; q, r)$ and the prime number theorem applied to $\pi(u)$: both have main term $\mathrm{li}(u)/\varphi(q)$ and $\mathrm{li}(u)$ respectively, and these cancel in $E$. Second, $|t| \le 1/(q\tau) = (\log N)^A/(qN)$ and $n \le N_0 = (\log N)^{A'}$ give $N|nt| \le (\log N)^{A+A'}/q$. ◻

*Remark* 11.10. It is worth keeping the Siegel–Walfisz saving in the form $e^{-c\sqrt{\log N}}$ rather than converting it to $(\log N)^{-C}$ for some fixed $C$. The assembly below multiplies the error by $\varphi(q) \le (\log N)^A$, sums it against $\sum_{n\le N_0} 1/n$, and compares it with $b/N_0 \gg N(\log N)^{-1-A'}$; a fixed power of $\log N$ would have to be chosen larger than $2(A + A') + 1$, which with $A = 11$, $A' = 12$ means $C > 47$. Carrying $e^{-c\sqrt{\log N}}$, which beats every power of $\log N$, removes the bookkeeping entirely.

**Lemma 11.11** (Annulus profile). *Let $q \le (\log N)^A$, let $\theta = 2\pi j/q + t$ with $(j, q) = 1$, and let $\Phi_q(x) = \sum_{n\ge1} \frac{(-1)^{n+1}}{n}\frac{c_q(n)}{\varphi(q)}(\mathrm{Re}\,\widetilde{I}(nx) - 1)$, where $c_q$ is the Ramanujan sum and $\widetilde{I}$ is the window of Appendix 12. Then*

$$\tfrac{1}{b}\log|G_B(\theta)| \;=\; \log M(q) + \Phi_q(N|t|) + o(1),$$

*an* equality*, where $\Phi_q(x) \to 0$ as $x \to 0$ and $\Phi_q(x) \to -\log 2 - \log M(q)$ as $x \to \infty$, so the profile interpolates between the peak height and the deep-minor bound of Section 9.*

*Proof.* In (1) write $\theta = 2\pi j/q + t$ and split the inner sum by residue class, $S_B(n\theta) = \sum_{r \bmod q} e(nrj/q)\sum_{p\equiv r} e(pnt)$. The classes with $(r, q) > 1$ contain at most $\omega(q)$ primes in total, so they contribute $O(\omega(q)) = O(\log\log N)$. To the remaining $\varphi(q)$ classes apply Lemma 11.9; the errors add to $\varphi(q)\cdot O(Ne^{-c\sqrt{\log N}}(\log N)^{A+A'}) \ll Ne^{-c\sqrt{\log N}}(\log N)^{2A+A'} = o(b/N_0)$, and the main terms assemble as

$$S_B(n\theta) = \frac{c_q(nj)}{\varphi(q)}\sum_{p\in B} e(pnt) + o(b/N_0) = \frac{c_q(n)}{\varphi(q)}\, b\,\widetilde{I}(Nnt) + o(b/N_0),$$

using $(j,q)=1$ to replace $c_q(nj)$ by $c_q(n)$. Summing over $n \le N_0$ against $1/n$ multiplies the error by $O(\log N_0)$ and leaves $o(b)$. The tail $n > N_0$ is the Abel summation of step 4 of Section 9, and the points where the series must be truncated are counted by Lemma 13.1. What remains is $-b\log 2 + b\sum_{n\ge1} \frac{(-1)^{n+1}}{n}\frac{c_q(n)}{\varphi(q)}\operatorname{Re}\widetilde{I}(Nnt)$, which is $b(\log M(q) + \Phi_q(N|t|))$ by the definition of $\Phi_q$ and the cyclotomic identity $-\log 2 + \sum_{n\ge1}\frac{(-1)^{n+1}}{n}\frac{c_q(n)}{\varphi(q)} = \log M(q)$ at $t=0$. □

*Remark* 11.12 (How the profile may and may not be used). Lemma 11.11 is an asymptotic equality, and the $o(1)$ is two-sided: Remark 11.15 finds the measurement above the profile at about half the sampled points. To use it as an upper bound one needs $\Phi_q(x) \le -c < 0$ for $x$ beyond the reach of the envelope. Writing $\Phi_q(x) = R_q(x) - \log(2M(q))$ with $R_q(x) = \sum_{n\ge1}\frac{(-1)^{n+1}}{n}\frac{c_q(n)}{\varphi(q)}\operatorname{Re}\widetilde{I}(nx)$, the bound $|\operatorname{Re}\widetilde{I}| \le \min(1, 2/x)$ of Lemma 12.2 together with $|c_q(n)| \le \varphi(q)$ gives $|R_q(x)| \le \frac{\pi^2}{3x}$, hence

$$\Phi_q(x) \le -\tfrac{1}{2}\log(2M(q)) \quad \text{for} \quad x \ge \frac{2\pi^2}{3\log(2M(q))},$$

which is $x \ge 12.0$ for $q=6$, $x \ge 16.4$ for $q=10$ and $x \ge 19.0$ for $q=4$. The envelope of Proposition 7.1 reaches only $x = N|t| \approx 1.5$. *The compact range in between is covered by neither.* It is a much smaller gap than the one it replaces — by Lemma 11.13 the quantity involved is a running average of an explicit elementary function, with no arithmetic in it — so a rigorous finite computation settles it; but we do not claim it. Numerically $\Phi_6$ is $-0.144$ at $x=1.5$, $-0.314$ at $x=2$ and below $-0.549$ from $x=3$ onwards, so the profile does not return to zero. We record the statement as Problem 11.3.

**Lemma 11.13** (Closed form of the profile)**.** *For $q \ge 3$ and $x > 0$,*

$$\Phi_q(x) = \frac{1}{\varphi(q)}\Big(\frac{1}{x}\int_0^x \log|\Phi_q(-e^{-iv})|\,dv - \log|\Phi_q(-1)|\Big),$$

*where $\Phi_q$ on the right is the $q$-th cyclotomic polynomial. In particular $\Phi_q(x)$ is a running average of* $\log|\Phi_q|$ *along the unit circle, started at the point $-1$ and normalised to vanish at $x=0$.*

*Proof.* $\operatorname{Re}\widetilde{I}(nx) = \sin(nx)/(nx) = \frac{1}{x}\int_0^x \cos(nv)\,dv$, so interchanging the sum and the integral,

$$\Phi_q(x) + \log(2M(q)) = \frac{1}{x}\int_0^x \sum_{n\ge1}\frac{(-1)^{n+1}}{n}\frac{c_q(n)}{\varphi(q)}\cos(nv)\,dv.$$

Writing $c_q(n) = \sum_{(r,q)=1}\cos(2\pi nr/q)$, pairing $r$ with $-r$, and using $\sum_{n\ge1}\frac{(-1)^{n+1}}{n}\cos(n\vartheta) = \log|2\cos(\vartheta/2)|$, the inner sum equals $\varphi(q)^{-1}\sum_{(r,q)=1}\log|2\cos(v/2+\pi r/q)|$. Finally $\prod_{(r,q)=1}(1+z\zeta_q^r) = (-z)^{\varphi(q)}\Phi_q(-1/z)$ gives $\sum_{(r,q)=1}\log|2\cos(v/2+\pi r/q)| = \log|\Phi_q(-e^{-iv})|$ for $|z|=1$, and the constant is the value at $v=0$. □

*Remark* 11.14 (Why $\Phi_q \le 0$ is false in general). Lemma 11.13 settles the shape of the question and also shows that the naive form of it is wrong. By Jensen's formula the mean of $\log|\Phi_q|$ over the whole circle is log of the Mahler measure of $\Phi_q$, which is 0. So if $|\Phi_q(-1)| = 1$ — equivalently $M(q) = 1/2$ — the running average starts at 0 and returns to 0, hence must be positive somewhere, and $\Phi_q \le 0$ fails. Numerically $q = 12, 15, 20, 21, 24$ are all counterexamples, with maxima up to $+0.109$. This costs nothing: those are exactly the $q$ with the smallest peak, and $\log M(q) + 0.109 = -0.584$ is still far below $\log(\sqrt{3}/2) = -0.144$.

It is the reason Problem 11.3 is stated for $\log M(q) + \Phi_q$ rather than for $\Phi_q$ alone. For $q = 6$, where $\Phi_6(-1) = 3$, the statement specialises to

$$\frac{1}{Y}\int_0^Y \log|1 + 2\cos v|\,dv \;\le\; \log 3 \qquad (Y > 0),$$

which is sharp only as $Y \to 0^+$: numerically the supremum over $Y \ge 1$ is 0.981.

*Remark* 11.15 (Numerical test of Lemma 11.11)*.* The profile is a prediction with no free parameters, so it can be falsified. We swept $t$ at $q = 6$ for $k = 40, 60, 100, 140$ and compared the measured drop $\frac{1}{b}\log|G_B(2\pi/6+t)| - \frac{1}{b}\log|G_B(2\pi/6)|$ against $\Phi_6(Nt)$, computing the window exactly from $B$ rather than from its smooth approximation, so that only the decoupling of equidistribution from the $t$-oscillation is being tested. Over 36 values of $x = Nt$ in $[0.25, 40]$:

| $b$ | 39 | 59 | 99 | 139 |
|---|---|---|---|---|
| max\|error\| | 0.107 | 0.082 | 0.054 | 0.059 |
| RMS error | 0.0354 | 0.0370 | 0.0202 | 0.0186 |
| $\sqrt{b}\cdot$ RMS | 0.221 | 0.284 | 0.201 | 0.219 |

The last row is flat, so the error is of the square-root size $b^{-1/2}$ one would expect, and it is an error in both directions — the profile is an asymptotic equality, not a pointwise majorant, and the measurement exceeds it at about half the sample points. This is why the $o(1)$ in Lemma 11.11 cannot be dropped. The saturation value is measured at $-0.52$ against the predicted $-\log 2 - \log M(6) = -0.549$, and at $x = 0.25, 1, 1.75$ the agreement at $b = 139$ is $-0.00308$ versus $-0.00310$, $-0.05233$ versus $-0.05221$, $-0.18862$ versus $-0.18683$.

*Remark* 11.16 (The notch at $q = 10$, and why it is a free test)*.* For $d = 2$ one has $5 \in B$, so $G_B(2\pi/10) = 0$ exactly by Lemma 3.1. The vanishing is *pointwise*: it removes one factor at one point and leaves the surrounding peak intact. Lemma 11.11, applied to $B \setminus \{5\}$ and multiplied by the explicit factor $|\cos(5\theta/2)| = |\sin(5t/2)|$, predicts the height of the two maxima flanking the notch with no free parameter. Maximising $b^{-1}\log|\sin(5t/2)| + \frac{b-1}{b}(\log M(10) + \Phi_{10}(Nt))$ gives

| $b$ | 39 | 59 | 99 | 139 |
|---|---|---|---|---|
| measured | $-0.446$ | $-0.404$ | $-0.366$ | $-0.351$ |
| predicted | $-0.429$ | $-0.394$ | $-0.361$ | $-0.345$ |
| $\log M(10) - \frac{3}{2}(\log b)/b$ | $-0.432$ | $-0.394$ | $-0.360$ | $-0.344$ |

The third row is the closed form obtained by balancing $b^{-1}\log(5t/2)$ against the quadratic part of $\Phi_{10}$; it agrees with the full profile to three decimals. So the flank height tends to $\log M(10)$ at the rate $\frac{3}{2}(\log b)/b$, and since $M(10) = 5^{1/4}/2 > 1/\sqrt{2} = M(4)$, the second-largest peak family for $d = 2$ is asymptotically $q = 10$ rather than $q = 4$. It remains far below $\sqrt{3}/2$.

*Remark* 11.17 (A sub-product bound, and its range)*.* One can also try to widen the certified envelope directly, by applying Proposition 7.1 to the sub-product over $\{a \in B : a \le A_1\}$: its factors have zeros spaced $2\pi/A_1$ apart, and $|G_B|$ never exceeds it, so the only cost is the smaller curvature $V_0(A_1) = \sum_{a \le A_1} a^2/4$. The counting decides where this is useful. An annulus exists only for $q \lesssim (\log N)^A$, and there the admissible cutoff is $A_1 \lesssim qN/(\log N)^A$, leaving $\pi(A_1) \approx bq/(\log N)^A$ factors. For $q$ near the top of the band this is a positive proportion of $b$ and the argument works; for small $q$ — including $q = 6$, the case that matters — the exponent collapses by a factor $(\log N)^A$ and no bound of the form $(\mathrm{const})^b$ survives. We record the device for the thin band where it applies and rely on Lemma 11.11 elsewhere.

**Problem 11.18** (Assembly)**.** Combine these to obtain $\varepsilon_d(k) \ll (\sqrt{3}/2)^k$, and hence $\varepsilon_k \to 0$ at a geometric rate.

We emphasise one negative result of the companion paper: there is *no* window analogue of the modulus-4 obstruction. Explicit odd sequences with $\varepsilon_2 = 0$ exist at every size, so no lower bound for $\varepsilon_d$ can be extracted from the aggregate residue-class spread. The route to Problem 11.18 must go through the upper bounds above.

## 12 Appendix: the window function

Lemma 11.11 needs one elementary object: the Fourier transform of the interval $[0, N]$, normalised to 1 at the origin.

**Definition 12.1.** $\widetilde{I}(x) = \dfrac{1}{N}\displaystyle\int_0^N e\Big(\frac{ux}{2\pi N}\Big)du = \frac{e^{ix}-1}{ix}$, extended by $\widetilde{I}(0) = 1$.

**Lemma 12.2.** *$\widetilde{I}$ is continuous, $\widetilde{I}(0) = 1$, $\operatorname{Re}\widetilde{I}(x) = \sin x/x$, and*

$$|\widetilde{I}(x)| = \frac{2|\sin(x/2)|}{|x|} \le \min\Big(1, \frac{2}{|x|}\Big), \qquad \operatorname{Re}\widetilde{I}(x) = 1 - \frac{x^2}{6} + O(x^4).$$

*Proof.* $\int_0^N e^{iux/N}du = N(e^{ix}-1)/(ix)$; then $|e^{ix}-1| = 2|\sin(x/2)| \le \min(|x|, 2)$, and the expansion is that of $\sin x/x$. □

*Remark* 12.3. The primes are not uniform on $[0, N]$ but have density $1/\log u$, so the window that actually occurs is $\widetilde{I}_{\log}(x) = b^{-1}\int_2^N e^{iux/N}\,du/\log u$. The second mean value theorem gives $|\widetilde{I}_{\log}(x)| \le \min(1, C\log N/|x|)$, which is the same shape with a $\log N$ in the constant; since $\Phi_q$ is only ever used through $|\widetilde{I}| \le \min(1, C/x)$, nothing downstream changes. In the numerical test of Remark 11.15 we did not use either approximation: the window was computed exactly as $b^{-1}\sum_{p\in B} e^{ipt}$, so that the only thing being tested was the decoupling of equidistribution from the oscillation in $t$.

## 13 Appendix: a self-contained one-interval Erdős–Turán bound

Step 3 of Section 9 needs to count the points of a finite family that land in one short interval. The Erdős–Turán inequality does this uniformly over all intervals, at the price of the Beurling–Selberg extremal function. For a single interval one can construct the majorant from the Fejér kernel in half a page, which is what we do, so that the section incurs no external reference.

Let $F_M(t) = \sum_{|n|\le M}(1 - \frac{|n|}{M+1})e(nt) = \frac{1}{M+1}\big(\frac{\sin\pi(M+1)t}{\sin\pi t}\big)^2$ be the Fejér kernel: a nonnegative trigonometric polynomial of degree $M$ with $\int_0^1 F_M = 1$.

**Lemma 13.1.** *Let $x_1, \dots, x_b \in \mathbb{R}/\mathbb{Z}$, let $I$ be an interval of length $2\delta$, and let $M \ge 1$. Put $S(n) = \sum_{j\le b} e(nx_j)$. Then*

$$\#\{j : x_j \in I\} \;\le\; 4b\Big(\delta + \frac{1}{M+1}\Big) + 4\sum_{n=1}^{M}\min\Big(2\delta + \frac{2}{M+1}, \frac{1}{\pi n}\Big)\,|S(n)|.$$

*Proof.* Write $w = 1/(M+1)$. For $w \le |t| \le \frac{1}{2}$ one has $\sin\pi t \ge 2t$, so $F_M(t) \le \big(4(M+1)t^2\big)^{-1}$ and hence $\int_{w\le|t|\le 1/2} F_M \le \frac{1}{2(M+1)w} = \frac{1}{2}$; consequently $\int_{|t|<w} F_M \ge \frac{1}{2}$.

Let $J$ be the interval with the same centre as $I$ and length $2\delta + 2w$, and set $P = 2\,(\mathbf{1}_J * F_M)$. Then $P$ is a trigonometric polynomial of degree at most $M$. If $x \in I$ then $x - J \supseteq (-w, w)$, so $(\mathbf{1}_J * F_M)(x) = \int_{x-J} F_M \ge \frac{1}{2}$ and $P(x) \ge 1$; thus $P \ge \mathbf{1}_I$ pointwise. Its coefficients are

$\widehat{P}(n) = 2\,\widehat{\mathbf{1}_J}(n)(1 - \frac{|n|}{M+1})$, and $|\widehat{\mathbf{1}_J}(n)| = |\frac{\sin(\pi n|J|)}{\pi n}| \le \min(|J|, \frac{1}{\pi|n|})$, while $\widehat{P}(0) = 2|J| = 4\delta + 4w$. Therefore

$$\#\{j : x_j \in I\} \le \sum_{j\le b} P(x_j) = \sum_{|n|\le M} \widehat{P}(n)\,\overline{S(n)} \le b\widehat{P}(0) + 2\sum_{n=1}^{M} |\widehat{P}(n)|\,|S(n)|,$$

which is the stated bound. □

*Remark* 13.2. The constant 4 is wasteful — the Beurling–Selberg majorant replaces it by 1 — but Section 9 only needs $o(b)$, and the bound is applied with $\delta = \delta_0 \to 0$ and $M = N_0 \to \infty$, so absolute constants are immaterial. The numerical values of the resulting bound are tabulated in Remark 9.7.

# 14 Appendix: the companion paper's Problem on random comparison

The companion paper asks whether $\mathrm{lm}_{P_k}/\mathrm{lm}_{R_k} \to 0$, where $P_k$ is the first $k$ odd primes and $R_k$ a random odd sequence of the same length drawn from the same range, and reduces this to comparing $\Gamma(P_k)$, which converges, with $\mathbb{E}[\Gamma(R_k)]$. The expectation is exactly computable.

**Theorem 14.1.** *Let $R_k$ be a uniformly random $k$-subset of the $N$ odd integers in $[3, p_k]$, where $N = (p_k - 3)/2 + 1$, and put $c = (N+1)/(k+1)$. Then*

$$\mathbb{E}[\Gamma(R_k)] = (1 - 2^{-k}) + 2c(2 - (k+2)2^{-k}).$$

*In particular $\mathbb{E}[\Gamma(R_k)] = 1 + 4c + O(k2^{-k})$, and since $c \sim p_k/(2k)$, this diverges as soon as $p_k/k \to \infty$.*

*Proof.* Identify the odd integers in $[3, p_k]$ with $\{1, \dots, N\}$ via $a = 2i+1$. For a uniform $k$-subset the order statistics satisfy $\mathbb{E}[X_{(j)}] = j(N+1)/(k+1)$ exactly, so $\mathbb{E}[a_{(j)}] = 1+2jc$. Hence $\mathbb{E}[\Gamma] = \sum_{j=1}^{k}(1+2jc)2^{-j} = (1-2^{-k})+2c(2-(k+2)2^{-k})$, using $\sum_{j\le k} j2^{-j} = 2-(k+2)2^{-k}$. □

*Remark* 14.2. No prime number theorem is needed: only $p_k/k \to \infty$, i.e. that the primes have density zero, which is elementary. Monte-Carlo agreement with Theorem 14.1 is within one standard error at $k = 100, 400, 800$ ($z = -1.46, -0.38, +1.07$ over 20,000 samples).

For a statement in probability rather than in mean one needs a lower tail bound, and the following suffices.

**Proposition 14.3.** *Let $R_k$ be a uniform $k$-subset of $\{1, \dots, N\}$ and let $X_{(J)}$ denote its $J$-th order statistic. For every integer $J \ge 1$ and every $\delta > 0$,*

$$\mathbb{P}[X_{(J)} \le \delta N/k] \;\le\; \frac{1}{\sqrt{2\pi J}}\Big(\frac{e\delta}{J}\Big)^{J}.$$

*Proof.* Write $m = \lfloor \delta N/k \rfloor$. The event $X_{(J)} \le \delta N/k$ says that at least $J$ of the $k$ sampled points lie in $\{1, \dots, m\}$. Bounding the probability of that event by the union over all $J$-subsets of $\{1, \dots, m\}$ of the event that the whole subset is sampled,

$$\mathbb{P}[X_{(J)} \le \delta N/k] \;\le\; \binom{m}{J}\frac{k(k-1)\cdots(k-J+1)}{N(N-1)\cdots(N-J+1)} \;\le\; \binom{m}{J}\Big(\frac{k}{N}\Big)^{J},$$

the last step because $(k-i)/(N-i) \le k/N$ for $0 \le i < J \le k \le N$. Now $\binom{m}{J} \le m^J/J!$ and, by Stirling, $J! \ge \sqrt{2\pi J}(J/e)^J$, so

$$\mathbb{P}[X_{(J)} \le \delta N/k] \le \frac{1}{\sqrt{2\pi J}}\Big(\frac{emk}{JN}\Big)^{J} \le \frac{1}{\sqrt{2\pi J}}\Big(\frac{e\delta}{J}\Big)^{J},$$

using $m \le \delta N/k$. □

**Corollary 14.4.** $\Gamma(P_k)/\Gamma(R_k) \to 0$ *in probability.*

*Sketch.* On the complement of the event above, $\Gamma(R_k) \geq 2^{-J}a_{(J)} \geq 2^{-J}(2\delta N/k)$. Fix $\delta = 1/2$ and let $J \to \infty$ slowly; the failure probability $(e/2J)^J/\sqrt{2\pi J}$ tends to 0, while $2^{-J}N/k \to \infty$ because $N/k \asymp \log k$ and $J$ may be taken as slowly growing as one likes. Since $\Gamma(P_k)$ is bounded, the claim follows. The quantifier bookkeeping is routine and is the only part not written out here. □

*Remark* 14.5. Proposition 14.3 was checked over 48 combinations $k \in \{100, 400, 1600, 3200\}$, $J \in \{1, 2, 4, 8\}$, $\delta \in \{0.25, 0.5, 1\}$ with $10^4$ samples each; it held in every case. Numerically the smaller quantity $(e\delta/2J)^J$ also dominated the observed frequencies throughout, but this reflects only the slack in the bound and not a sharper theorem: the expected number of sample points below $\delta N/k$ is $\delta$, not $\delta/2$, so no factor 2 is available at this step. The stated form is the one the argument proves, and the conclusion is unaffected in any case, since $J$ is chosen after $\delta$.

## Data availability

Every numerical value quoted in this paper was produced by a script in the accompanying repository, and each script writes a log file of the same name. The scripts, the logs and the Lean development are available at `https://github.com/mathlab0911/arithmetic-landscape`.

## References


[AR14] K. Alyahya and J. E. Rowe, *Local optima and weight distribution in the number partitioning problem*, in: Parallel Problem Solving from Nature — PPSN XIII, Lecture Notes in Computer Science **8672**, Springer, 2014, pp. 862–871.

[BE56] P. T. Bateman and P. Erdős, *Monotonicity of partition functions*, Mathematika **3** (1956), 1–14.

[BMOR18] M. A. Bennett, G. Martin, K. O'Bryant and A. Rechnitzer, *Explicit bounds for primes in arithmetic progressions*, Illinois J. Math. **62** (2018), no. 1–4, 427–532. arXiv:1802.00085.

[DRZZ25] A. Dong, N. Robles, A. Zaharescu and D. Zeindler, *Strange and pseudo-differentiable functions with applications to prime partitions*, Res. Number Theory **11** (2025), 52. arXiv:2412.20102.

[Hel12] H. A. Helfgott, *Minor arcs for Goldbach's problem*, preprint (2012), arXiv:1205.5252.

[FF98] F. F. Ferreira and J. F. Fontanari, *Probabilistic analysis of the number partitioning problem*, J. Phys. A **31** (1998), 3417–3428.

[Ku05] A. V. Kumchev, *An invitation to additive prime number theory*, Serdica Math. J. **31** (2005), 1–74; open access. Lemma 3.1 is the bound we use.

[Sr25] P. Srivastav, *Log-free bounds on exponential sums over primes*, preprint (2025), arXiv:2505.07803.

[MB26] M. V. N. Murthy and M. Brack, *Quantum density of states and integer partitions: a semiclassical approach*, preprint (2026), arXiv:2607.06146.

[RS54] K. F. Roth and G. Szekeres, *Some asymptotic formulae in the theory of partitions*, Quart. J. Math. Oxford (2) **5** (1954), 241–259.

[SHF03] P. F. Stadler, W. Hordijk and J. F. Fontanari, *Phase transition and landscape statistics of the number partitioning problem*, Phys. Rev. E **67** (2003), 056701.

[V08] R. C. Vaughan, *On the number of partitions into primes*, Ramanujan J. **15** (2008), 109–121.